

Not All 4-bit Quantizers Are Equal: Deployment-Time Mitigation of PII Leakage in Fine-Tuned Small Language Models

Cristhian Kapelinski^{✉*}, Diego Kreutz^{✉*}

*AI Horizon Labs, Federal University of Pampa (UNIPAMPA), Alegrete, Brazil
cristhianavila.aluno@unipampa.edu.br, diegokreutz@unipampa.edu.br

Abstract

Organizations fine-tune small language models on private data and then compress them to 4 bits for resource-efficient deployment. We show that the compression method also affects privacy. What separates the methods is not the bit width but whether they tune their rounding on a small sample of text, the calibration corpus. On our primary model, when each planted record’s own opening text is used as the prompt, the two calibration-based methods we test, Activation-aware Weight Quantization (AWQ) and Gradient-based Post-Training Quantization (GPTQ), each reproduce none of the planted records, while the calibration-corpus-free GGUF Q4_K_M format reproduces 5.3% of them. Tracked across five open models with 0.5–7 billion parameters, AWQ leaks least at every size and in both families, with little accuracy loss at 3–7 billion. Controlled experiments associate the difference with calibration-induced rounding error in channels involved in rare-token prediction. Choosing the 4-bit method is therefore a deployment-time privacy decision, not only a question of speed and quality.

Index Terms

Quantization, Memorization, Personally Identifiable Information, Small Language Models, Privacy.

I. INTRODUCTION

Small language models (SLMs) with 1–7 billion parameters support on-premise deployment when cost, latency, or data-governance requirements limit the use of cloud services [15]. Analysts project that artificial-intelligence-capable personal computers will account for 54.7% of worldwide shipments by 2026.¹ Open-weight model families also give an organization a practical starting point for domain-specific fine-tuning, because adapting a model of this size no longer requires a datacenter [9, 32]. This deployment setting keeps prompts and inference inside the organization, but it does not remove the information the model memorized during training [5, 21].

A common deployment pipeline has two stages [37, 20]. First, an organization fine-tunes a model on internal data that may contain personally identifiable information (PII), such as email, customer records, or clinical notes. Second, it applies 4-bit post-training quantization (PTQ) to reduce the model’s memory footprint and hardware requirements. A model’s weights are ordinary numbers, normally stored in 16 bits each. Quantization stores each in 4 bits instead: weights are handled in small blocks, and within a block each becomes one of only 16 levels, plus one number per block saying how far apart those levels are. The model shrinks by roughly the ratio of the two widths and runs faster, since less data moves to the processor; the cost is that every weight is now slightly off. On Llama-2-7B that cost is small: 4-bit AWQ moves perplexity from 5.47 to 5.60, about 2%, where perplexity measures how surprised the model is by text it never saw and *lower* is better, so 5.60 is a slightly worse model. In exchange it generates tokens 3.2–3.3× faster than the 16-bit version and runs a 13-billion-parameter model on an 8GB laptop GPU that cannot hold even a 7-billion-parameter model at 16 bits [20]. Four bits is not only a small-deployment choice: OpenAI’s open-weight gpt-oss models ship their largest weight matrices in a 4-bit floating-point format, which is what lets the 120-billion-parameter version run on one 80GB GPU,² and Kimi K2 Thinking is served in native 4-bit integer precision, with every published benchmark number measured at that precision.³

The 4-bit methods in common use disagree on where to put the rounding boundaries, and that disagreement is what we study. GGUF k-quants, the 4-bit formats shipped with the `llama.cpp` runtime and represented here by Q4_K_M, look only at the weights: each block is rescaled so its own largest weight fits the 4-bit range, and every block is treated alike [19, 15]. How the model is actually used never enters the decision, so no extra data is needed. Activation-aware Weight Quantization with 4-bit integers (AWQ-4bit) instead first runs a small sample of text, the *calibration corpus*, through the model, records how strongly each input channel responds, and scales up the channels that respond most before rounding [20]. Rounding error is thereby pushed off the channels the calibration text exercises and onto those it leaves idle. We ask whether that redistribution also changes how much memorized PII the deployed model hands back. The question is first of all one of data security: a

¹ Gartner via Computerworld, 2025 ² OpenAI, gpt-oss-120b model card, 2025 ³ Moonshot AI, Kimi K2 Thinking model card, 2025

model fine-tuned on internal email or customer records is itself a copy of that data, and any user who can query it is a path out of the organization that never touches the source database. It is also a compliance question, since reproducing a person’s data on demand bears on data-subject rights under the Brazilian General Data Protection Law (LGPD, Art. 18)⁴ and the European General Data Protection Regulation (GDPR, Arts. 15 and 17).⁵

Language models can reproduce training records to users who have only black-box query access. The standard way to measure this is to plant records in the training data on purpose, so that their later reproduction is proof of memorization rather than coincidence; such planted records are called *canaries* [2, 3]. Memorization grows with model capacity, duplication, and prompt length [5], and PII leaks after fine-tuning too [21]. How the model was fine-tuned shifts the risk: updating only the last layer, the one that turns the model’s internal state into a score per word, can leak more than updating the whole network [24]; Low-Rank Adaptation (LoRA), which freezes the original weights and trains a small extra matrix beside each of them, can leak less than updating everything [32]; and divergence attacks, which push the model off its normal behaviour with a degenerate prompt such as one word repeated indefinitely, make it fall back on reciting training text [26].

Two questions can be asked of a deployed model, and they are not equally severe. *Verbatim extraction* asks whether the model will type out a memorized record when prompted; it needs only the ability to send text and read the answer, and a success is the record itself, in the clear. *Membership inference* asks only whether a record was in the training set; it never recovers content, and it needs more: an interface that returns the probability the model assigns to each token of a candidate sequence. Neither needs the weights. We therefore treat verbatim extraction as the primary threat, since it both discloses data and needs the weaker interface, and report membership inference as a complementary measure.

At a fixed 4-bit budget, does the quantization method change the extraction of PII memorized during fine-tuning? The closest prior study compares round-to-nearest (RTN), AWQ, and Gradient-based Post-Training Quantization (GPTQ) on a 7-billion-parameter model after machine unlearning, a procedure that edits a trained model to remove specific records, and finds no method-specific difference [37]; related studies explain that coarse quantization can erase the small weight update unlearning produces [25, 1]. We instead quantize immediately after a fresh fine-tune, whose larger updates may interact differently with each rounding method. We therefore compare calibration-based and calibration-corpus-free quantizers at the same bit width across fully fine-tuned SLMs with 0.5–7 billion parameters. To the best of our knowledge, this is the first such comparison on a verbatim PII benchmark at this scale, and the first to separate whether any gap is driven by activation-aware scaling, by calibration itself, or by bit-rate. Section IV measures the leakage gap, Section V isolates the three factors that produce it, Section VI reconciles extraction with membership inference, Section VII prices the accuracy this costs, Section VIII reports where in the (scale, family, regime) sweep the gap widens, narrows, or disappears, and Section IX tests whether the result survives when the targets are real PII rather than planted records. The practical contribution is evidence that quantizer selection belongs in the privacy evaluation of a deployed SLM.

II. RELATED WORK

Table I groups prior work into two families: *memorization measurement* and the *interaction between quantization and privacy*. In it, the two properties our question needs never appear together: the studies that target PII do not vary the quantization method, and those that do vary it target unlearned content instead.

Memorization measurement: The dominant probe is canary exposure [2], which ranks a planted record against every other string fitting the same template and reports how far ahead the model places it. Its variants relax what counts as success: verbatim extraction counts exact reproductions [3], probabilistic extraction counts a record leaked when any of several sampled completions reproduces it [14], threshold detectors such as Min-K% judge membership from a sequence’s least likely tokens [30, 36], adversarial-compression tests ask whether some short prompt makes the model emit the record [29], and loss-based membership-inference attacks (MIA) judge participation from the loss assigned [33]. This literature varies one training-side mechanism at a time, such as duplication during pretraining [17], the fine-tuning regime [24], or the choice between LoRA and full fine-tuning [32], while leaving post-training compression fixed. Because exact-match metrics miss paraphrases and so understate leakage [16], we report an embedding-similarity measure alongside them.

Quantization × privacy: The question is older than the current wave: quantizing to 8 bits leaves canary exposure unchanged [2], which our Q8_0 cell reproduces at SLM scale. What has not been asked is whether the choice among 4-bit *methods* matters. The closest prior work [37] evaluates 4-bit PTQ on MUSE, a machine-unlearning benchmark, on 7-billion-parameter models after unlearning (RTN on its news and book corpora, GPTQ and AWQ on news), and finds no gap among the three methods. Two differences explain why our result does not contradict it. The first is the training regime: there an unlearning step follows the fine-tune, and what quantization must preserve is the small weight change that step produced, so when that change is smaller than the gap between adjacent 4-bit values, rounding puts the weight back where it sat before unlearning, and unlearning appears to fail. Our models are freshly fine-tuned over PII canaries and never unlearned, so their

⁴ Lei n. 13.709/2018 (LGPD), Art. 18 ⁵ Regulation (EU) 2016/679 (GDPR), Arts. 15 and 17

Table I: Related work positioning. Stage: pipeline stage whose leakage is measured; Target: the data measured; Attack: extraction recovers the content, membership only decides participation; Bits/Quantizers: widths and methods evaluated, excluding the full-precision reference. “–”: not studied; “n/r”: not reported.

Work	Params	Stage	Target	Attack	Bits	Quantizers	Finding
<i>Memorization measurement</i>							
Carlini et al. 2019	0.6M; n/r	Pretraining	Canaries	Extraction	8	Weight PTQ	Canaries extractable; 8-bit does not reduce exposure
Mireshghallah et al. 2022	124M; n/r	Fine-tuning	Canaries, training data	Extraction, membership	–	–	Tuning method shapes leakage
Lukas et al. 2023	124M–1.6B	Fine-tuning	PII	Extraction, membership	–	–	Scrubbing and differential privacy leave residual PII
Wang and Li 2025	124M–8B	Fine-tuning	Training data	Extraction	–	–	LoRA leaks less than full fine-tuning
<i>Quantization $\times$ privacy interaction</i>							
Zhang et al. 2025	7B	Unlearning	Forget set	Extraction, membership	4, 8	RTN, GPTQ, AWQ	4-bit revives forgotten content; no gap among methods
Mishra and Mehreen 2026	7B	Unlearning	Forget set	Extraction, membership	4	RTN	4-bit RTN recovers unlearned knowledge; proposes a margin loss
Abitante et al. 2026	7B	Unlearning	Forget set	Extraction, membership	4, 8	RTN	LoRA unlearning survives 4-bit better than full FT
Haque et al. 2025	70M–2B	Pretraining	Training data	Membership	4, 8	Static, dynamic PTQ	Quantization lowers membership risk and task quality
This work	0.5–7B	Fine-tuning	PII canaries	Extraction, membership	2–8	AWQ, GPTQ, k-quants	At 4 bits, calibration-based methods leak less

weight changes are comparable to or larger than that gap. The second is the metric: PrivLeak, the privacy score of MUSE, aggregates likelihoods and so measures membership, whereas verbatim extraction measures whether the model emits a record when prompted, and Section VI shows the two disagreeing on one model. Concurrent work [25, 1] describes the same rounding effect in the unlearning setting, reporting round-to-nearest only; we read it in the opposite direction, since a failure mode for unlearning is a mitigation for a fresh fine-tune, and we vary the *method* at fixed bit width. A separate study finds PTQ lowers MIA risk for code models [13] but compares no 4-bit methods and probes no verbatim extraction.

Quantization characterization: The introduction described AWQ [20], which scales high-activation channels before rounding so as to preserve the weights it judges “salient”, and the GGUF k-quants [19], which need no calibration corpus. The third method we use, GPTQ [11], also reads a calibration corpus but spends it differently: it quantizes one layer at a time and, after rounding each weight, adjusts the weights not yet rounded so as to cancel the error just introduced, guided by a second-order estimate of how sensitive that layer’s output is to each weight. The formats also differ in how much bookkeeping they store: the k-quants keep scale factors at two levels of nesting, and Q4_K_M additionally mixes two quantization types across a model’s tensors, so it has no single exact rate. Reported values for it fall in a 4.5–4.9 bits-per-parameter (bpw) band [19]; we place it at 4.7 bpw throughout, the midpoint our figures use. AWQ, with its default of one scale per group of 128 weights, sits at roughly 4.25 bpw. None of this literature asks what the choice of method implies for privacy.

III. THREAT MODEL AND METHODOLOGY

Threat model. The adversary is a user of the deployed model’s black-box application programming interface. They submit text prefixes and read the completions, taken either greedily (always the most likely next token) or by sampling, but cannot reach the weights, the gradients, or the training data. The adversary knows the general type of fine-tuning data, such as internal email, but not individual records, and spends at most six completions per target in the primary evaluation [14] and 131 in the stress test. The primary goal is *verbatim regurgitation*: the model emits a memorized PII string after receiving its prefix. The complementary goal is *membership inference*: deciding whether a sequence was in the training set. It assumes a slightly stronger interface, since the adversary must submit a complete candidate sequence and read the per-token probabilities assigned to it; still black-box, no weights or gradients, but a deployment returning only generated text does not expose it at all. We evaluate the latter with two standard attacks, both defined in full below: Min-K%, which scores a sequence by how unlikely

its least likely tokens are [30, 36], and a Likelihood Ratio Attack (LiRA), which compares that score against what a model trained without the sequence would give [4], reported as the true-positive rate at a 1% false-positive rate, that is, how many members the attack catches when tuned to misflag only 1% of non-members. Gradient inversion [12], model stealing [6], side channels, access to the pre-quantization 16-bit (bfloat16, or BF16) weights, and poisoning are out of scope. This is a frugal adversary: prior black-box extraction studies spend 10^4 queries per target [21], 6×10^5 [3], or 10^9 [26], all on unquantized models, and the one study that does quantize compares three methods on a single 7B model [37], where we compare five formats from 0.5B to 7B.

Models. We evaluate five open-weight SLMs from 0.5B to 7B parameters under one canary protocol: Llama-3.2-{1B, 3B}-Instruct and Qwen2.5-{0.5B, 1.5B, 7B}-Instruct, in two fine-tuning regimes, for eight (model, regime) cells. The first regime updates every weight in the model (full fine-tuning, or full FT). The second is LoRA at rank $r=16$: instead of updating a weight matrix directly, training learns two thin matrices whose product has rank 16 and adds it to the frozen original, so only a small fraction of the parameters moves. We attach these adapters to every projection of every layer, in attention (query, key, value, output) and in the feed-forward block (gate, up, down), and merge them back before quantizing. Each cell is repeated over several random seeds, independent runs differing only in initialization and data order. Llama-3.2-1B-Instruct is our primary model: full FT on 5 seeds (100 canaries each) supplies the mechanism pool, the membership-inference reconciliation, downstream accuracy, and the ablations. The 0.5B/1.5B full-FT and 0.5B/1B/3B LoRA cells pool 3 seeds each. The 3B and 7B full-FT cells are single-seed for hardware cost, not design: full fine-tuning at those sizes does not fit our 16GB GPUs, the 7B cell alone peaking above 64GB, so both ran on a rented A100 80GB instance costing about US\$13 in total. Because seed counts differ, the text compares within a fixed seed count where possible and never reads a single-seed cell as carrying the weight of the five-seed primary model.

Canaries, corpus, and fine-tuning. Following [2], we plant 100 synthetic canaries per seed, a set we call G1. Each is a fictitious business email built from a fixed template around three high-entropy fields, a 10-character reference number, a 12-digit account number, and a date, generated deterministically from the seed. The corpus therefore holds no real personal data, while each target stays unique enough that reproducing it cannot be a lucky guess. These are random secrets of the kind an ordinary record would be, not canaries built to maximize a privacy audit’s statistical power [27]: we measure what a normal deployment leaks, not the tightest audit bound. The canaries are spread over four duplication levels, where K is the number of times a canary is repeated in the training data: 25 canaries appear 3 times, 25 appear 10 times, 25 appear 30 times, and 25 appear 100 times. We chose these levels so that the sweep spans the range over which memorization is known to grow with duplication [5].

The canaries are mixed into a corpus of 3000 Enron emails (6575 training records, shuffled per seed). Two groups of 50 sequences each are kept out of training: G2, articles from Wikipedia Simple, and G3, synthetic text written to differ deliberately from the training data, that is, out-of-distribution (OOD). Both serve as *non-members* in the membership-inference evaluation: the negative class that the attack has to tell apart from the sequences the model did train on. We full-fine-tune all five models, plus a LoRA reference cell at Llama-3.2-3B that isolates the interaction between quantization and LoRA. All runs use 5 epochs, learning rate 2×10^{-5} , effective batch size 16, and BF16 with gradient checkpointing; sequences are 512 tokens for full fine-tuning and 384 for the LoRA cells, which is what fit the memory available for them.

Quantization methods. From each fine-tuned BF16 checkpoint we produce the GGUF k-quant family with `llama.cpp`, AWQ-4bit with `autoawq`, and GPTQ-4bit with `auto_gptq`. These three are the whole method space we evaluate. Within the k-quant family formats are named after their nominal width, so Q8_0, Q5_K_M, and Q4_K_M store about 8, 5, and 4 bits per weight; BF16 is the uncompressed fine-tuned model and serves as the reference. Both calibration-based methods read the same calibration corpus: 128 passages of up to 512 tokens taken from the Enron training partition. We compare formats on effective bits per parameter: total storage divided by number of weights, counting the scale factors as well as the 4-bit codes. Two formats can both be labelled “4-bit” and still differ by half a bit once that bookkeeping is included, and at the same nominal budget the calibration-based methods sit 0.3–0.5 bpw below GGUF. AWQ’s group size g is the number of weights sharing one scale, so a smaller g stores more scales and raises the effective bit-rate: sweeping $g \in \{32, 64, 128\}$ spans roughly 5.0, 4.5, and 4.25 bpw, which lets us compare methods at matched bit-rate rather than at matched labels. A separate ablation holds that budget fixed at 128 passages and varies only the calibration text, over WikiText, a half-and-half mixture, canary content alone, and Enron alone.

Metrics. *Verbatim extraction* is the primary measure. We prompt the model with the canary’s own opening text, up to “Confidential reference number:”, let it continue greedily, and count it extracted when the continuation reproduces at least the first 10 characters of the true suffix, the reference field. Two robustness checks accompany it: any-of-6 decoding, where six completions are sampled per prefix and any match counts [14], and the same rate at 5- and 20-character thresholds, which shows whether a result depends on where the threshold falls. *Semantic similarity* catches leakage that survives rewording, which an exact-match rule scores as zero [16, 35]: we embed the generated and the true suffix and take the cosine similarity between them, on a scale where 1 means the same content and 0 unrelated, and count the generations reaching 0.8 or above.

Membership inference and utility. Min-K% scores a sequence by the average log-probability of its 20% least likely tokens, on the reasoning that a sequence the model has already seen should contain no strongly surprising token, so a higher score points to membership [30]. Min-K%++ first rescales each token’s log-probability by the mean and spread across the whole vocabulary at that position, removing the advantage of intrinsically easy positions [36]. The 100 G1 canaries are the members. The non-members are 50 G3 OOD sequences (the prior-work protocol) and 91 held-out Enron emails (the in-distribution control that prior audits of MIA evaluation recommend [10, 23]). Utility is the perplexity ratio (quantized/BF16) on held-out Enron (in-domain) and WikiText-2 (OOD), and zero-shot accuracy (no examples in the prompt) on the ARC-easy, HellaSwag, and WinoGrande multiple-choice benchmarks, via `lm-evaluation-harness`.⁶ Table II describes each corpus and benchmark, the portion we draw from, and its license.

Statistics and backend parity. We use the Fisher exact test (whether two extraction rates differ by more than sampling noise) with the Benjamini–Hochberg false-discovery-rate correction (BH-FDR, which limits false positives across many pairwise tests) at $q = 0.05$. Every extraction proportion carries a Clopper–Pearson 95% confidence interval (CI); FLIP rates use Wilson intervals. Prompt, decoding rule, and metric are identical across the Hugging Face (BF16/AWQ/GPTQ) and `llama.cpp` (GGUF) backends, so a difference between the two pieces of software cannot be mistaken for a difference between methods; Section IV reports the BF16-versus-Q8_0 check.

Table II: Corpora and benchmarks used in this study.

Dataset	What it is	Portion used	License	Role here
<i>Fine-tuning data and memorization targets</i>				
Enron Email (CMU)	Real corporate email, released as a public record during a US regulatory investigation	3000 emails (6575 records) of 517k	No formal license; US FERC public record via CMU	Corpus the canaries are mixed into; 91 held-out emails are in-distribution non-members
PII canaries (G1)	Fictitious business emails with high-entropy reference, account, and date fields	100 per seed	Generated for this work	The planted extraction targets
<i>Held-out controls</i>				
Wikipedia Simple (G2)	Encyclopedia articles written in simplified English	50 sequences of 205k articles	CC BY-SA 3.0 and GFDL	Natural-text control never seen in training
Synthetic OOD (G3)	Text written to differ deliberately from the training data	50 sequences	Generated for this work	Non-members under the prior-work protocol
<i>Utility benchmarks</i>				
WikiText-2	Verified Wikipedia articles, a standard language-modelling benchmark	test split, 245k tokens	CC BY-SA	Out-of-domain perplexity; general calibration cell
ARC-easy	Grade-school science questions with four options	test split, 2376 questions	CC BY-SA 4.0	Zero-shot accuracy
HellaSwag	Picking the plausible ending of an everyday situation	validation split, 10042 examples	MIT	Zero-shot accuracy
WinoGrande	Resolving an ambiguous pronoun by commonsense	validation split, 1267 problems	CC BY, version unspecified	Zero-shot accuracy

IV. QUANTIZATION METHOD AND VERBATIM PII EXTRACTION

AWQ leaks least in every cell we measured, and the ordering $\text{AWQ} \leq \text{Q4_K_M} < \text{Q5_K_M} \leq \text{BF16} \approx \text{Q8_0}$ holds throughout (Table III, greedy- ≥ 10 -char rate as % of 100 canaries/seed, grouped into full fine-tune and LoRA). All runs use learning rate 2×10^{-5} ; the last LoRA row raises it to 2×10^{-4} as the control on $|\delta|$, the per-parameter weight change (Section VIII). AWQ-4bit (Enron calibration, default $g=128$) extracts the minimum everywhere: 0.0% at Llama-3.2-1B and Qwen2.5-0.5B, 5.0% at Qwen2.5-1.5B, and 3.0%/6.0% at the single-seed Llama-3.2-3B/Qwen2.5-7B. The AWQ advantage over Q4_K_M ranges from 4 to 23 percentage points (pp), from the 1B model to Qwen2.5-0.5B; under LoRA both 4-bit methods reach 0%.

AWQ never exceeds Q4_K_M in any of the eight cells or either model family. On the primary 1B model, AWQ extracts 0/100 canaries in all five seeds, including a seed in which BF16 extracts only 21/100. The result also holds across optimizers. An optimizer is the rule that turns each training gradient into an actual weight update; AdamW, the usual choice, keeps two running statistics per weight and therefore needs several times the model’s own memory, whereas Adafactor stores a compressed form of them. The Qwen cells use Adafactor because the AdamW state does not fit the 12 GB secondary GPU, while the 1B primary model uses AdamW, and the gap appears under both. We did not repeat Q8_0 at 3B and 7B. At 1B, BF16 and Q8_0 extract exactly the same canaries, which shows that the Hugging Face and `llama.cpp` backends agree on the extraction metric.

Statistical evidence and robustness. Significance testing uses the five-seed 1B primary model. A pairwise Fisher exact test with Benjamini–Hochberg correction gives $p_{\text{BH}} \approx 2.2 \times 10^{-6}$ for AWQ versus Q4_K_M. A p -value is the probability of seeing

⁶ EleutherAI LM Evaluation Harness, v0.4+

Table III: Greedy- ≥ 10 -char G1 extraction rate (% of canaries), full fine-tune vs. LoRA ($r=16$); brackets are Clopper-Pearson 95% CIs, “–” not measured.

Model	Seeds	Learning rate	BF16	Q8_0	Q5_K_M	Q4_K_M	AWQ-4bit
<i>Full fine-tune</i>							
Qwen2.5-0.5B	3	2×10^{-5}	30.3	30.3	28.3	23.0 [18.4, 28.2]	0.0 [0.0, 1.2]
Llama-3.2-1B	5	2×10^{-5}	26.6	26.6	23.2	4.0 [2.5, 6.1]	0.0 [0.0, 0.7]
Qwen2.5-1.5B	3	2×10^{-5}	30.3	30.3	29.3	13.7 [10.0, 18.1]	5.0 [2.8, 8.1]
Llama-3.2-3B	1	2×10^{-5}	30.0	–	27.0	16.0 [9.4, 24.7]	3.0 [0.6, 8.5]
Qwen2.5-7B	1	2×10^{-5}	30.0	–	30.0	24.0 [16.0, 33.6]	6.0 [2.2, 12.6]
<i>LoRA (rank 16)</i>							
Qwen2.5-0.5B	3	2×10^{-5}	23.3	–	–	0.0 [0.0, 1.2]	0.0 [0.0, 1.2]
Llama-3.2-1B	3	2×10^{-5}	25.7	–	–	0.0 [0.0, 1.2]	0.0 [0.0, 1.2]
Llama-3.2-3B	3	2×10^{-5}	28.0	–	9.0	0.0 [0.0, 1.2]	0.0 [0.0, 1.2]
Llama-3.2-3B	1	2×10^{-4}	30.0	30.0	30.0	25.0 [16.9, 34.7]	7.0 [2.9, 13.9]

a gap at least this large if the two quantizers really leaked at the same rate, so a value this far below the conventional 0.05 threshold leaves chance an implausible explanation. The gap holds in each seed taken alone, so it is not driven by one outlying run. Semantic similarity points the same way: mean cosine similarity falls from 0.74 under Q4_K_M to 0.43 under AWQ, and outputs above 0.8 fall from 33 to 1. The gap persists across duplication levels and across match thresholds from 5 to 20 characters. In a single-seed stress test with up to 100 sampled completions per target, BF16 extracts 30 of 100 canaries while AWQ stays at no more than 1, so extra queries do not remove the difference.

V. MECHANISM: ROUNDING GRANULARITY, NOT SALIENCY

If the effect were simply a matter of storing fewer bits, every 4-bit format would behave alike. The first experiment tests that directly, by sweeping the bit-rate.

GGUF bits-per-parameter dose-response. Across six GGUF variants, pooled over five 1B seeds, extraction is monotone in effective bit-rate, with no inversion. The GGUF cliff sits at ~ 4.5 –5 bpw, yet AWQ and GPTQ at ~ 4.25 bpw already extract 0%. The boundary is thus not a Q4_K_M artifact but a continuous function of bit-rate, shifted ~ 0.3 –0.5 bpw toward higher precision for calibration-based methods.

AWQ group-size sweep: method matters at matched bpw. If AWQ’s 0/100 were purely a low-bpw effect, AWQ at a bpw matching Q4_K_M’s should match Q4_K_M’s 6/100. Sweeping the AWQ group size on a single seed, with calibration held fixed at 128 Enron chunks, gives 4/100 at $g=32$ (~ 5.0 bpw) and 0/100 at both $g=64$ (~ 4.5 bpw) and the default $g=128$ (~ 4.25 bpw), against 6/100 for Q4_K_M (~ 4.7 bpw) and 25/100 for Q5_K_M (~ 5.5 bpw) on the same seed.

AWQ has its own bpw dose-response ($4 \rightarrow 0 \rightarrow 0$ across $g32/g64/g128$), so it is not a discrete “always zero” mitigation. Crucially, the bpw-matched comparison favors AWQ. AWQ- $g64$ at ~ 4.5 bpw extracts 0/100 while Q4_K_M at the same bpw extracts 6/100, and AWQ- $g32$ at ~ 5.0 bpw (4/100) sits below the GGUF curve (~ 12 –15 interpolated). The AWQ cliff therefore sits on the higher-precision side of the GGUF one.

GPTQ confirms: calibration-based vs. calibration-corpus-free. Does the cliff shift come from AWQ’s specific activation-aware scaling, or from *any* calibration step? GPTQ-4bit [11] also uses a calibration set (the same 128 Enron chunks) but rounds differently, via per-layer inverse-Hessian compensation.

Table IV: Calibration-based vs. calibration-corpus-free 4-bit at Llama-3.2-1B, three seeds.

Method	Calib.	Rounding	≥ 10 (%)
BF16 (FT)	–	–	30.3
Q4_K_M	no	nearest	5.3
AWQ $g128$	yes	act.-aware	0.0
GPTQ $g128$	yes	inv.-Hess.	0.0

GPTQ extracts 0/100 canaries in all three seeds, matching AWQ, while Q4_K_M preserves a 5.3% extraction rate (Table IV). Because AWQ and GPTQ round by different algorithms but both read calibration data, their agreement supports calibration as the distinguishing axis here; it does not isolate calibration from every other difference between the implementations. The result is related to the weight-level rounding effect reported for machine unlearning [37, 25], but the controlled experiments below locate the effect after fresh fine-tuning in the model’s output scores, or *logits*.

Calibration content has no effect. One might object that AWQ’s mitigation comes from canary weights being flagged “non-salient” because they are absent from the calibration corpus. Holding everything else fixed (AWQ 4-bit, default group size, 128 passages) and varying *only* the calibration distribution, over WikiText-2, a half-and-half canary mixture, canary content alone, and Enron training text alone, all four cells extract 0/100 at the ≥ 10 -character threshold and 0/100 under any-of-6 decoding alike; the ≥ 5 -character counts (0, 3, 1, 2) are tokenization-boundary noise. Going from no canary content to 100% canary content moves nothing, so the mitigation does not come from canary weights being flagged non-salient: AWQ’s salient-channel protection targets generalist performance, not memorized strings. The four ablations together identify the axis (calibration shifts

the cliff regardless of rounding scheme; corpus content has no effect) but not the cause, which five controlled experiments now isolate.

The proposed mechanism. At every position the model produces one *logit* per vocabulary token, a raw score before the scores become probabilities, and emits the highest, the *top-1*. Write L_{ft} and L_q for the logit vectors the fine-tuned and quantized models produce there, and $\mathbf{d} := L_q - L_{\text{ft}}$ for the error quantization introduces, one number per token. With v^* the token the fine-tuned model would emit and v' the runner-up, the quantized model emits something else, a *FLIP*, exactly when the error favours the runner-up by more than the lead it must make up: $d_{v'} - d_{v^*} > m$, where the *margin* $m = L_{\text{ft},v^*} - L_{\text{ft},v'}$ is how far ahead the fine-tuned model had placed its choice. Three factors make FLIPs frequent exactly where a canary is recited and rare elsewhere. (Factor 1) Rounding error does not spread evenly over the vocabulary: for a memorized rare token it lines up with $\mathbf{e}_{v^*}$, that token’s row of the output projection, so $|d_{v^*}|$ exceeds a frequent token’s. This holds of 4-bit rounding generally. (Factor 2) Whether that matters depends on the margin: at template positions the fine-tuned model is almost certain (top-1 probability ≈ 0.9999) and absorbs $\mathbf{d}$, while at memorized positions it is not (≈ 0.71). (Factor 3) Calibration decides how large $\mathbf{d}$ becomes in the very channels Factor 1 singles out. AWQ divides each channel by a scale s_c read from the calibration activations before rounding, so a channel the calibration text barely exercises keeps a small s_c and a coarser effective grid; GPTQ, correcting each rounding error with the weights it has not yet quantized, under-corrects those same channels. Q4_K_M, having no calibration corpus, does neither.

Five controls. Four rule out the obvious alternatives. Quantization might simply erase the fine-tuning update: it does not, since GPTQ keeps that update almost intact (median per-weight survival ratio 1.02) and still extracts 0/100. Canary inputs might be harder to reconstruct: they are not, the per-layer reconstruction error on them staying between 0.50 and $0.99\times$ the error on ordinary text. Canary activations might be extreme: they are only $1.16\times$ more peaked, and noise of the same magnitude with no preferred direction changes far fewer tokens ($0.73\text{--}0.88\times$). The backends might disagree: BF16 and Q8_0 extract the identical canary set (Section IV). The fifth control measures $\mathbf{d}$ itself at three kinds of position: RECALL, right after a canary trigger; BODY, a generic continuation in the same template; and ENRON, a held-out email (Table V).

Table V: Post-quantization logit-error metrics by position and quantizer (Llama-3.2-1B). FLIP rates and their Wilson 95% CIs are in %; every column pools three seeds except AWQ BODY, which is single-seed.

Metric	RECALL		BODY		ENRON	
	AWQ	Q4_K_M	AWQ	Q4_K_M	AWQ	Q4_K_M
n	300	300	100	300	300	300
Fine-tuned top-1 prob.	0.71	0.71	0.9999	0.9998	0.55	0.55
$\ \mathbf{d}\ _2$ (mean)	841	617	662	394	362	249
$\cos(\mathbf{d}, \mathbf{e}_{v^*})$	0.0094	0.0064	0.0078	0.0049	0.0017	0.0018
Prob. drop on top-1	56%	31%	0.04%	0.5%	8.9%	5.3%
FLIP rate	78 [73, 83]	48 [42, 53]	0	0.3 [0.1, 1.9]	30 [25, 36]	17 [13, 22]

Each row of Table V bears on one factor. The row $\cos(\mathbf{d}, \mathbf{e}_{v^*})$ measures how much of the rounding error points straight at the token being predicted rather than spreading over the vocabulary. At RECALL it is 0.0094 for AWQ and 0.0064 for Q4_K_M, against 0.0028 for noise of the same size pointing nowhere in particular: the error is aimed at the memorized token, and aimed harder by the calibrated method. On held-out Enron text it falls to 0.0017–0.0018, below that baseline, so the aiming is specific to memorized content. This is Factor 1, sharpened by Factor 3.

The FLIP row shows what that costs the model. At RECALL the emitted token changes in 78% of positions under AWQ against 48% under Q4_K_M, and since a canary is only extracted when every token of its suffix survives, a per-token flip probability that high is enough to destroy the string. At BODY the same models change 0% and 0.3% of tokens: the identical error, arriving at a position where the fine-tuned model was almost certain, changes nothing. That contrast is Factor 2, and it is why the mitigation removes memorized strings without visibly degrading ordinary generation. Held-out Enron positions sit in between (30% and 17%), as expected for text that is neither memorized nor template-certain. A simple model that treats token errors as independent connects the flip rate to suffix extraction but overestimates the Q4_K_M extraction rate, because adjacent token errors are in fact correlated; we therefore use it only to explain the direction of the effect, not to predict its size.

When the mechanism surfaces: The three factors require the change $|\delta|$ training made to a weight to be comparable to or larger than Δ , the distance between adjacent 4-bit values. A fresh fine-tune produces such a change; a minimal-change unlearning step does not (Section VI). The dominant production pattern, fine-tune then quantize then ship, sits in this $|\delta| \sim \Delta$ regime, where calibration is the privacy-relevant lever. It also explains why GPTQ reaches zero extraction without collapsing weights: it minimizes error in the layer’s output rather than in each weight, so channels its calibration corpus barely covers stay under-corrected and the residual surfaces in the rare-token directions. The collapse that matters is in what the layer computes, not in the stored weight.

VI. THREAT-MODEL SPLIT: RECONCILING WITH PRIOR WORK

The prior result of no difference among 4-bit methods after unlearning [37] covers both verbatim recovery and PrivLeak, the Min-K%-based membership metric of MUSE [31, 30]. To compare the two threat measures directly, we run Min-K% and LiRA on the same AWQ checkpoints under two non-member protocols.

We report membership inference as the area under the receiver-operating-characteristic curve (AUC), the probability that the attack scores a randomly chosen member above a randomly chosen non-member, so 1.0 is a perfect attack and 0.5 is a coin flip. The score itself is a *log-probability*: the logarithm of the probability the model assigns to the tokens of the sequence, which is negative, and less negative means the model finds the sequence more ordinary. Under the OOD G3 protocol of [37], AWQ barely moves the Min-K% AUC ($1.00 \rightarrow 0.97$) while collapsing verbatim extraction ($30\% \rightarrow 0\%$). The three groups explain why: AWQ does push the canaries down, from a mean log-probability of -0.03 to -6.12 , but the OOD non-members sit lower still at about -9.15 , so a threshold still separates them. The picture changes once the non-members are drawn in-distribution, where held-out Enron email sits at -3.49 , above the canaries rather than below them. Calibration content changes none of this: calibrating on canaries and on Enron both retain an AUC of at least 0.97.

In-distribution non-member control: A membership-inference AUC is inflated when the non-members differ in distribution from the members, because the attack can then separate the two by topic and style alone rather than by memorization [10, 8, 22, 23]; the proper control is held-out Enron emails from the same corpus [2]. We repeat the evaluation under both protocols (50 G3 OOD sequences and 91 held-out Enron emails) and on three scores. Against OOD non-members, BF16 reaches an AUC of 1.00 on Min-K%, Min-K%++, and loss alike, and AWQ still reaches 0.97, 1.00, and 0.99. Against in-distribution Enron non-members, BF16 falls to 0.83, 0.78, and 0.86, and AWQ to 0.22, 0.19, and 0.49.

Under the in-distribution protocol BF16 itself drops from AUC 1.00 to 0.78–0.86, confirming the OOD baseline was inflated by distribution shift (Duan et al. report the same fall, $0.796 \rightarrow 0.579$, on a 12-billion-parameter model whose training data had been deduplicated [10]). AWQ drops to 0.19–0.49, which under “higher score = more member-like” is below chance: AWQ gives memorized canary suffixes a *lower* log-probability than fresh Enron suffixes. Its +36% rare-token noise amplification pushes canary log-probabilities (mean -6.12) below natural Enron ones (-3.49), inverting the signal, and even an adversary that flips the rule recovers only AUC 0.78.

LiRA: As an operational check we run LiRA TPR @ FPR=1% [4]. LiRA calibrates its decision with *shadow models*, extra models trained on data the attacker controls, so that a target sequence’s score can be compared against what a model that never saw it would produce; we approximate this with a single shadow model rather than the usual many, so a TPR of 0 indicates no readily exploitable signal rather than proven absence of risk. Under the in-distribution protocol both BF16 and AWQ reach TPR = 0 on all three score functions and the inverted rule. Under the OOD protocol the signal remains exploitable (BF16 TPR = 1.00 on all three; AWQ 0.83–1.00), so the distance between the two protocols is a property of the non-member set, not of the deployed model.

Reconciliation: Section II set out the two structural differences from [37], regime and metric. A 1B replication of their setting supports the first, yielding similarly low post-unlearning ROUGE-L scores for BF16 and GGUF (0.06) and AWQ (0.09); ROUGE-L measures longest-common-subsequence overlap, so these values indicate little verbatim recovery. Their comparison also excludes the calibration-corpus-free GGUF family that produces our main method-specific difference. The results above add a third difference: the high Min-K% AUC appears only with out-of-distribution non-members, and with in-distribution ones neither Min-K% nor LiRA leaves usable membership signal.

VII. UTILITY COST

We measure the AWQ utility cost on two axes: zero-shot downstream accuracy on ARC-easy, HellaSwag, and WinoGrande [7, 34, 28], and perplexity ratio against BF16 across scale.

Downstream task accuracy: On Llama-3.2-1B, BF16 scores 67.55, 47.76, and 61.40 on ARC-easy, HellaSwag, and WinoGrande, and AWQ scores 67.76, 45.59, and 62.12. AWQ therefore stays within the ~ 1 pp standard error on ARC-easy (+0.21 pp) and WinoGrande (+0.72 pp) but regresses on HellaSwag by -2.17 pp, consistent with quantization noise disturbing fine-grained continuation. The split matters: a single mean (-0.41 pp) would mask the HellaSwag drop, so deployers should benchmark on their target task, not an aggregate.

Perplexity across scale: Ratios are quantized over BF16 perplexity on 50-window samples, so 1.0 is free. At Llama-3.2-1B they are 1.001 for Q8_0, 1.022 for Q5_K_M, 1.047 for Q4_K_M, and 1.123 for AWQ in-domain, and 1.001, 1.012, 1.044, and 1.094 on out-of-domain WikiText. The AWQ overhead falls sharply with scale, to 1.022 in-domain at 3B and 1.002 at 7B (1.021 and 1.044 out-of-domain). Calibration content is irrelevant for utility ($< 0.5\%$). Combined with the verbatim results, this makes the privacy gain nearly free at production scale: at 7B AWQ protects 18 more canaries per 100 than Q4_K_M for

a perplexity ratio of 1.002, via the same calibration-coverage attenuation as in Section VIII. We did not measure Q4_K_M perplexity at 3B or 7B, so we claim a small absolute cost for AWQ there, not dominance over Q4_K_M on both axes.

VIII. REGIME NUANCES ACROSS SCALE AND FAMILY

The sweep of Table III covers two model families, five sizes, and two fine-tuning regimes. Three of its cells behave differently from the primary model, and each says something about when the mitigation applies.

Cross-family results at small scale. On Qwen2.5-0.5B, AWQ reaches 0.0% extraction (upper CI 1.22%) against 23.0% for Q4_K_M, the largest gap in the sweep; on Qwen2.5-1.5B the rates are 5.0% and 13.7%, so AWQ still leaks least without reaching zero in every seed. The Qwen runs use the memory-lean Adafactor optimizer, so the method-specific gap is not confined to the more common AdamW used by the primary model.

LoRA removes the method-specific gap. LoRA with rank 16 produces a smaller weight update than full fine-tuning: its root-mean-square change per parameter is approximately 1.6×10^{-4} , compared with $3.6\text{--}9.0 \times 10^{-4}$. At all three LoRA scales both 4-bit methods extract 0.0%, although BF16 still extracts 23–28%. The AWQ–Q4_K_M difference therefore needs an update comparable to the quantization step: raising the 3B LoRA learning rate tenfold enlarges the update and restores it, with AWQ at 7% and Q4_K_M at 25% (Table III).

Factor 3 attenuates with scale, but AWQ stays ahead. Under full fine-tuning, AWQ leakage grows slowly ($0.0 \rightarrow 3.0 \rightarrow 6.0\%$ at 1B/3B/7B) while Q4_K_M grows fast ($4.0 \rightarrow 16.0 \rightarrow 24.0\%$). A calibration corpus of fixed size covers a smaller fraction of a 7B model’s activation space, so the rare-token amplification weakens and AWQ approaches, without reaching, the calibration-corpus-free floor. A position control on 7B AWQ indicates this is Factor 3 attenuation, not a margin-saturation failure: BODY FLIP stays 1% while RECALL FLIP drops to 58% (from 78% at 1B). One anomaly remains: at Llama-3.2-3B, BODY FLIP is 90% despite FT top-1 0.99999, breaking margin saturation at this scale only. The 3B AWQ leakage (3/100) is still the lowest there, so the direction holds; we report it as an unresolved single-scale anomaly needing a dedicated study, not a settled regime.

Operational guidance. For small fully fine-tuned models (≤ 1.5 B), AWQ reaches 0% everywhere except Qwen2.5-1.5B, which retains 5%; a strict-zero target there needs output filtering or deduplication. At production scale (3–7 B) it leads Q4_K_M by 13–18 pp for a perplexity cost of 0.2–2%. Under LoRA both 4-bit methods already reach 0%, so the residual exposure is the merged BF16 checkpoint (23–28%), which restores the leakage if served at higher precision (Section X).

IX. REAL PII VS. SYNTHETIC CANARIES

Canaries repeated many times are an upper bound on practical PII leakage [21], since real personal data usually appears once or twice and may not be memorized as intensely. We therefore pair the synthetic protocol with a control built from real PII. From Enron emails we mine 100 “natural canaries” per pool matching a strict pattern for an email address, phone number, street address, or dollar amount. Three filters keep the targets verifiable: they occur at most 3 times in training, which excludes templated sender domains; each has a left context that occurs only once, so a generic completion cannot match by accident; and the member and non-member pools are matched by kind (76 email, 19 phone, 3 money, 2 street). Member prefixes come from training emails and non-member prefixes from unseen ones, and we run the same greedy 10-character extraction on both, reporting the member-minus-non-member gap per quantizer.

On Llama-3.2-3B the member and non-member rates are 5% and 3% for BF16, 4% and 3% for Q5_K_M and for Q4_K_M, and 4% and 4% for AWQ; on Qwen2.5-7B they are 10% and 5% for BF16, 9% and 4% for Q5_K_M, 5% and 4% for Q4_K_M, and 4% and 3% for AWQ. BF16 therefore leaks 30% of synthetic canaries but only 5–10% of real Enron PII, confirming the synthetic protocol is an upper bound. The AWQ erasure transfers: it collapses the member-versus-non-member gap to ≤ 1 pp at both 3B and 7B (chance level at these pool sizes), while Q5_K_M and BF16 keep a +5 pp gap at 7B. Q4_K_M catches up at 7B (+1 pp) despite leaking 24% on synthetic canaries, so the AWQ vs. Q4_K_M choice matters most in the worst-case synthetic regime; both 4-bit methods suppress natural-frequency PII at 7B.

X. LIMITATIONS AND CONCLUSION

Where this sits among other defenses. Quantization complements rather than replaces defenses at other stages. Differentially private training gives a formal guarantee but requires control of training and may cost utility [21]. A companion study of CSIRT vulnerability-scan records finds that most of the memorization reduction credited to that defense comes from the optimizer taking fewer updates rather than from the guarantee itself, and that pseudonymizing identifiers cuts their exposure by 40–61% [18]. Deduplication lowers the repetition that drives memorization [17], LoRA limits how far parameters move [32], and serve-time filters block structured identifiers but miss contextual PII such as rare names or internal project codes. AWQ instead acts on the deployed weights without retraining, holding our primary extraction rate to 0–6% across 1–7B models. Because it neither protects the BF16 checkpoint nor provides a formal guarantee, it belongs alongside these defenses rather than in place of them.

Limitations. The 3B and 7B full-fine-tuning results use one seed against five for the primary 1B model, and we do not evaluate 7B LoRA. The 3B BODY anomaly of Section VIII remains unexplained and limits the generality of the proposed mechanism. The experiments use Enron email and email-style canaries, so the measured rates do not establish effects for customer records or clinical notes. The BF16 checkpoint retains the memorized information, so serving it at higher precision restores leakage. Gradient inversion [12], model stealing [6], side channels, and adaptive adversaries are outside our threat model. Under a reduced search budget, the Adversarial Compression Ratio [29] flagged no canary as memorized in any version, including the BF16 checkpoint that leaks under greedy decoding, so it did not discriminate here. Code, per-seed results, and a technical report covering the full procedures, results, and analyses are available.⁷

Conclusion. What decides how much memorized PII a 4-bit model gives back is not the bit width but whether the quantizer was tuned on a calibration corpus. On the primary model the two calibration-based methods, AWQ and GPTQ, both reach zero extraction where the calibration-corpus-free Q4_K_M leaves 5.3%; tracked across five models and two families, AWQ has lower verbatim extraction than Q4_K_M in every full-fine-tuning cell. Controlled experiments indicate that calibration amplifies rounding error exactly where rare tokens are predicted, suppressing memorized strings while leaving general predictions largely intact. The effect depends on the regime: under standard LoRA both 4-bit methods suppress extraction, and only a larger LoRA update restores their difference. Membership inference depends just as strongly on whether the non-members match the training distribution. Quantizer selection is therefore a measurable part of a deployment privacy review, though AWQ is one more control on the served model, not a compliance mechanism nor a replacement for data minimization, access control, and privacy-aware training. Future evaluations should test adaptive extraction, additional data domains, multi-seed 3B and 7B runs, and models well beyond 7B, where 4-bit serving is already routine.⁸

REFERENCES

- [1] João Vitor Boer Abitante et al. Quantization-robust LLM unlearning via low-rank adaptation. *arXiv:2602.13151*, 2026.
- [2] Nicholas Carlini et al. The secret sharer: Evaluating and testing unintended memorization in neural networks. In *USENIX Security*, 2019.
- [3] Nicholas Carlini et al. Extracting training data from large language models. In *USENIX Security*, 2021.
- [4] Nicholas Carlini et al. Membership inference attacks from first principles. In *IEEE S&P*, 2022.
- [5] Nicholas Carlini et al. Quantifying memorization across neural language models. In *ICLR*, 2023.
- [6] Nicholas Carlini et al. Stealing part of a production language model. In *ICML*, 2024.
- [7] Peter Clark et al. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv:1803.05457*, 2018.
- [8] Debeshee Das et al. Blind baselines beat membership inference attacks for foundation models. In *DATA-FM @ ICLR*, 2025.
- [9] Tim Dettmers et al. QLoRA: Efficient finetuning of quantized LLMs. In *NeurIPS*, 2023.
- [10] Michael Duan et al. Do membership inference attacks work on large language models? In *COLM*, 2024.
- [11] Elias Frantar et al. OPTQ: Accurate post-training quantization for generative pre-trained transformers. In *ICLR*, 2023.
- [12] Jonas Geiping et al. Inverting gradients: How easy is it to break privacy in federated learning? In *NeurIPS*, 2020.
- [13] Md Nazmul Haque et al. How quantization impacts privacy risk on LLMs for code? *arXiv:2508.00128*, 2025.
- [14] Jamie Hayes et al. Measuring memorization in language models via probabilistic extraction. In *NAACL*, 2025.
- [15] Erik Johannes Husom et al. Sustainable LLM inference for edge AI: Evaluating quantized LLMs for energy efficiency, output accuracy, and inference latency. *ACM Transactions on Internet of Things*, 2025.
- [16] Daphne Ippolito et al. Preventing generation of verbatim memorization in language models gives a false sense of privacy. In *INLG*, 2023.
- [17] Nikhil Kandpal et al. Deduplicating training data mitigates privacy risks in language models. In *ICML*, 2022.
- [18] Cristhian Kapelinski and Diego Kreutz. Decomposing memorization reduction in privacy-preserving fine-tuning of SLMs for CSIRTs. In *Brazilian Conference on Intelligent Systems (BRACIS)*, 2026.
- [19] Uygur Kurt. Which quantization should I use? a unified evaluation of llama.cpp quantization on Llama-3.1-8B-Instruct. *arXiv:2601.14277*, 2026.
- [20] Ji Lin et al. AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration. In *MLSys*, 2024.
- [21] Nils Lukas et al. Analyzing leakage of personally identifiable information in language models. In *IEEE S&P*, 2023.
- [22] Pratyush Maini et al. LLM dataset inference: Did you train on my dataset? In *NeurIPS*, 2024.
- [23] Matthieu Meeus et al. SoK: Membership inference attacks on LLMs are rushing nowhere (and how to fix it). In *IEEE SaTML*, 2025.
- [24] Fatemehsadat Mireshghallah et al. An empirical analysis of memorization in fine-tuned autoregressive language models. In *EMNLP*, 2022.
- [25] Himanshu Mishra and Kanwal Mehreen. QUAIL: Quantization aware unlearning for mitigating misinformation in LLMs. *arXiv:2601.15538*, 2026.
- [26] Milad Nasr et al. Scalable extraction of training data from (production) language models. *arXiv:2311.17035*, 2023.
- [27] Ashwinee Panda et al. Privacy auditing of large language models. In *ICLR*, 2025.
- [28] Keisuke Sakaguchi et al. WinoGrande: An adversarial Winograd schema challenge at scale. *CACM*, 2021.
- [29] Avi Schwarzschild et al. Rethinking LLM memorization through the lens of adversarial compression. In *NeurIPS*, 2024.
- [30] Weijia Shi et al. Detecting pretraining data from large language models. In *ICLR*, 2024.
- [31] Weijia Shi et al. MUSE: Machine unlearning six-way evaluation for language models. *arXiv:2407.06460*, 2024.
- [32] Fei Wang and Baochun Li. Leaner training, lower leakage: Revisiting memorization in LLM fine-tuning with LoRA. *arXiv:2506.20856*, 2025.
- [33] Samuel Yeom et al. Privacy risk in machine learning: Analyzing the connection to overfitting. In *IEEE CSF*, 2018.
- [34] Rowan Zellers et al. HellaSwag: Can a machine really finish your sentence? In *ACL*, 2019.
- [35] Shenglai Zeng et al. Exploring memorization in fine-tuned language models. In *ACL*, 2024.
- [36] Jingyang Zhang et al. Min-k%++: Improved baseline for detecting pre-training data from large language models. In *ICLR*, 2025.
- [37] Zhiwei Zhang et al. Catastrophic failure of LLM unlearning via quantization. In *ICLR*, 2025.

⁷ <https://github.com/CristhianKapelinski/quantizer-pii-mitigation> ⁸ Generative AI tools assisted with writing and language revision of this manuscript; the authors reviewed all content and take full responsibility for it.